Alaqsa Akbar

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EDUCATION

King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, Saudi Arabia **Sep 2022 – May 2026**
BSc in Computer Engineering (Research Concentration with a focus on AI) GPA: 3.92/4.00

Georgia Institute of Technology, Atlanta, GA, United States **Aug 2024 – Jan 2025**
Exchange student in Computer Engineering GPA: 4.0/4.0

Oxford University, Oxford, United Kingdom **Jul 2024 – Aug 2024**
Summer Program in Advanced Artificial Intelligence Grade: A

WORK EXPERIENCE

SDAIA-KFUPM Joint Research Center for AI, Dhahran, Saudi Arabia — ***Researcher*** **Aug 2025 – Present**
Advised by: Dr. Motaz Alfarraj

- Researched and developed a state-of-the-art video generation model for continuous sign language
- Utilized SLURM servers to train diffusion transformer models conditioned through retrieval augmented discrete pose data generated by vector quantized variational autoencoders

Baker Hughes, Houston, TX, USA – ***ML Research Intern*** **Jun 2025 – Aug 2025**
Advised by: Dr. Moradi Peyman

- Researched and developed an algorithm to extract low frequency data from distributed acoustic sensing using compressed sensing.
- Assisted in automation projects by documenting and designing test procedures of embedded systems architectures, and developing machine learning models for abnormality detection

Georgia Tech OLIVES Lab, Atlanta, GA, United States – ***Research Assistant*** **Sep 2024 – Dec 2024**
Advised by: Prof. Ghassan AlRegib

- Cleaned multiple datasets each worth terabytes of data on seismic information and optical diseases
- Managed datasets: formatted, documented, and uploaded to HuggingFace

SDAIA-KFUPM Joint Research Center for AI, Dhahran, Saudi Arabia — ***Researcher*** **Feb 2024 – Aug 2024**
Advised by: Dr. Abduljabbar Siddiqui

- Created a data pipeline to generate a dataset in real time throughout the model training phase using OpenCV
- Modified an adversarial attack library (Adversarial Robustness Toolbox) to train a customized adversarial patch attack model robust against adverse weather conditions by utilizing a project gradient descent (PGD) algorithm

RESEARCH

- **Akbar, A.** & AlFarraj, M. (2026) *SSLVG: Saudi Sign Language Video Generation Using Retrieval-Augmented Diffusion Transformers* **(In progress) (Undergraduate Thesis)**
- Allahham, A., & **Akbar, A.** (2025, May 2). *Re-Colonization of Architectural Knowledge through AI: an Echo Chamber*. 3rd Architecture and Design Educators Forum, Architecture and Design Commission. Dhahran, KSA. **(Conference)**
- Alshehri, I.*, Alshehri, A.*, Almalki, A.*, Bamardouf, M.*, & **Akbar, A**.* (2024, August 31). *BreachSeek: a Multi-Agent automated penetration tester.* arXiv.org. https://arxiv.org/abs/2409.03789 **(Preprint)**

*** indicates equal contribution**

EXTRA-CURRICULARS

Qimam Fellowship, Riyadh, Saudi Arabia – ***Fellow*** **Mar 2025 – Present**

- 50 students selected globally out of 18,000+ applicants to participate in a prestigious leadership mentoring program

KAUST Academy AI Specialization, Oxford, United Kingdom – ***Participant*** **Jan 2024 – Aug 2024**

- Chosen as one of 15 students from over 7000 applicants to attend summer courses at Oxford University
- Took courses on Deep Learning, Computer Vision, Natural Language Processing, and Deep Unsupervised Learning

Saudi Mathematics Olympiad Team, Thuwal, Saudi Arabia – ***Member*** **2017 – 2019**

- Represented Saudi Arabia at the Junior Balkan Mathematical Olympiad in Cyprus where Saudi Arabia got 3rd

RELEVANT TECHNICAL SKILLS/COURSEWORK

Computing Languages and Technologies – Python, Java, C/C++, Verilog, SLURM, Git, Linux
Coursework – Computer Vision, NLP, Deep Unsupervised Learning, Deep Learning, Data Science, Computer Architecture, Data Structures and Algorithms, Object Oriented Programming, Embedded Systems, Computer Networks

ADDITONAL SKILLS

Languages: Bilingual in Arabic and English (114 TOEFL)